The Evolution of Solar Electric Vehicles: Technologies, Barriers, and Research Frontiers

Alessandro Marino, Hiroshi Tanaka, Priya Ramanathan, Sofia Kuznetsova, Carlos Eduardo Vieira
Department of Energy Engineering, Politecnico di Milano, Italy

Abstract— **The depletion of conventional energy resources and escalating environmental challenges urgently require technological innovation in mobility. Hybrid and electric vehicles now present realistic alternatives to reduce fossil fuel dependence and limit emissions. This work provides a detailed review of solar electric vehicles (SEVs), focusing on the implementation of photovoltaic (PV) technologies as primary or auxiliary propulsion energy sources. Major advances in system architecture—spanning power electronics, energy management, and high-efficiency PV modules—are discussed alongside persistent barriers, including range limitations, cost structures, and grid integration complexities. The review identifies prominent areas of research such as advanced MPPT techniques, smart charging algorithms, and the optimization of PV panel design for vehicular integration. Critical knowledge gaps are highlighted, especially regarding large-scale deployment, the real-world performance of PV systems in mobile conditions, and the bidirectional impact on distributed energy systems. Ultimately, the findings emphasize that further interdisciplinary research and policy support are needed to realize the widespread adoption of SEVs as an essential element of sustainable transportation.**

Keywords — Battery, electric vehicle, MPPT, optimal energy management, solar vehicles, photovoltaic panels, super-capacitor.

I. INTRODUCTION

Nowadays, one of the main world issue is the gradual reduction of oil reserves. At the same time climate change concern is growing, then a key objective is the reduction of pollutants emission. In this context, transportation is a key sector despite that a huge reduction in emission is achieved with the new vehicles. Till now, there are some concerns over air quality and noise, especially in urban areas [1]. The aforesaid factors are also driving towards an increasingly diffusion of technologies based on renewable sources [2].

In the last years, the technological development has proposed a variety of innovative strategies to face this challenge, polling out new options as the hybrid powertrain system. The resulting hybrid electric vehicles (HEVs) use an electrical propulsion module in addition to the traditional combustion one. HEVs are equipped with an electrical energy storage system (ESS), an electric motor, and a transmission system that couples both the electric and the thermal engine. The additional electric motor allows savings on the thermal motor consumptions, and obviously lowers the emission of toxic substances. The HEV market share is increasing continuously [3], [4] and vehicles based on this technology are widespread with the aim to decarbonize the transport sector, involving both the public and private transportations.

The technological development has led to major improvements in the electric car environment, creating hybrid type marketable products that are very advanced, On the other hand, they are not yet fully competitive with traditional internal combustion engine (ICE) cars [5]. More specifically, HEVs are quite expensive, and the need of an increasing development of storage capacity in the ESS is nowadays one of the biggest designers' issue. To overcome range anxiety they require higher capacity and ancillary systems for recharging devices whose cost for large-scale diffusion is very substantial [6][7]. In many countries, Authorities are shaping a drastic constrain in terms of the car emission and it is expected that the restrictions will increase in the next years. This aspect allows to increase the spread of the electric vehicles against the fully thermal vehicles, *e.g.* the EU commission with regards to the project Horizon 2020 has provided some procedural guideline constraints in terms of emission of diesel engines. The price to adapt the combustion based cars during the years will be comparable with the purchase of a new electric car. Moreover, economic support policies are also becoming a common practice in such a direction with purchase subsidy and vehicle taxes with exemptions for the electric vehicles [8]. All these aspects provide different benefit to both the individuals and the society. A large penetration of HEVs will also lead to a great electric power demand that may strongly affect the power generation system as well as the transmission and distribution ones [9], [10]. Finally, it is worth to note that the end life of batteries goes against the use of HEV for the aim of pollution reduction [11], thus advanced battery recycling strategies are mandatory [12].

As known, the use of solar energy has many benefits because it is available, sustainable, renewable, and it is a clean energy source. The use of solar panels in case of a fuelefficient HEV enables to transfer these characteristics to the hybrid solar vehicles (HSVs) in order to ensure higher autonomy and a lower environmental impact.

The aim of this paper is to review the current HSV status and the development of the related technologies, the social impacts, and the potential brought by the solar field.

II. HYBRID SOLAR VEHICLE CONCEPT

Different HEVs architectures are widely discussed in literature: series, parallel, and parallel-series [13][14]. In the series configuration, the ICE is connected to an electric generator (EG) that recharges the ESS or a series of super capacitors (SC). The powertrain is made up by an electric motor (EM). This architecture has a drawback because the energy passes through a series of components (ICE, ESS, converter, EM). Each part has an own efficiency and consequently the overall powertrain global efficiency is reduced. This architecture allows controlling easily the power delivered to each wheel, and consequently the traction control is simplified. On the other hand, the parallel configuration connects both the EM and the ICE with a mechanical transmission (T). It is possible to find different applications of this hybrid system: in the first one, the ICE is the main component and the electric part works to help the combustion engine. Other vehicles mainly operate with the electrical part, and the thermal engine help electrical system. Usually a large ESS is required. The series–parallel configuration in a HEV combines the features of both series and parallel HEV. The key point is that a mechanical connection is used between ICE and T (in comparison with the series system), and an additional generator between ICE and the power converters (PC) is used (in comparison with the parallel system). This architecture is more complex than the previous ones, and it leads higher manufacturing cost. A great advantage of this structure is more flexibility in the control of the whole system. All these configurations are depicted in Figs. 1-2-3. The two latter architectures have been experimented in the last decades for many hybrid cars sold in the market. The series configuration is preferable [15] when a HSVs are designed, thanks to its simplicity.

A simply and a comprehensive scheme of the series architecture for the HSVs is discussed in [16] and a block scheme is depicted in Fig. 4. In such a case, the photovoltaic system (PV) is connected with the EG through a DC bus. These units recharge the ESS (battery array or super capacitor) both in driving and parking mode conditions. It is worth noting that the design of the DC-DC converters and their controllers are crucial for the control power regulation and the communication with the common DC bus because the highpower DC/DC converters are key elements in the HSVs. They are responsible for connecting the PV panels with the DC bus in the powertrain of the vehicle. Moreover, they regulate the voltage and current levels during the ESS charging period and control the energy exchange between the ESS and the motor DC/AC converter. In this configuration there are usually bidirectional converters. For example, the ESS can be also recharged by a regenerative braking mode of operation [17][19], then the DC/AC converter operates as a rectification stage. Obviously, the DC/DC converter designed for the PV modules is a unidirectional converter. The advantages and disadvantages of several topologies of DC-DC converters, are presented and compared in Section IV. A system enabling a traditional vehicle to achieve the advantages of both HEVs and HSVs is another practical solution. In this perspective, a mild solar hybridization of conventional vehicles is treated in [20].

III. IMPACT OF HYBRID SOLAR VEHICLE

Many researches have been carried out to investigate the impact on the environment, economy and network (*i.e.* in terms of energy production and delivery) of using the HSVs. The core findings of these effects will be discussed in detail in the following sections.

A. Environmental impact

Air pollution is an important issue in the metropolitan areas, and vehicle emission is one of the main reasons. The HEVs are claimed to be green and environmental-friendly since have drastically reduced the emission of CO_2. However, the HEVs use electricity generated from power plants for battery charging, and large part of the power generators is based on technologies that unfortunately produces greenhouse gas emissions. Therefore, a parameter called “wells-to-wheels" emissions has been proposed [21], [22] to take into account the emissions over the whole lifetime of the vehicle. The parameter considers the energy and materials used to power a

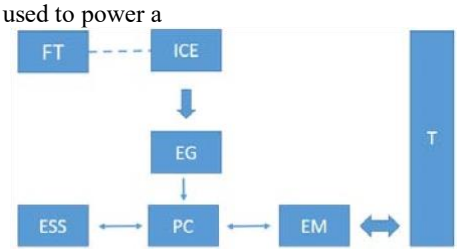

Fig. 1. Example of a series architecture for Hybrid Electric Vehicle. FT: fuel tank, PC: power converter, T: transmission.

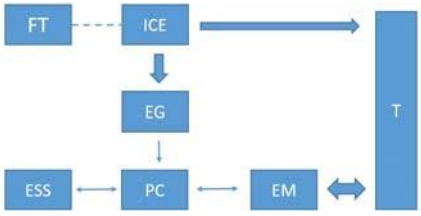

Fig. 2. Example of a series-parallel architecture for Hybrid Electric Vehicle.

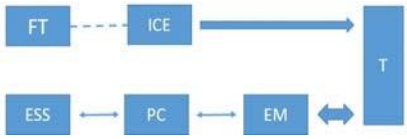

Fig. 3. Example of a parallel architecture for Hybrid Electric Vehicle.

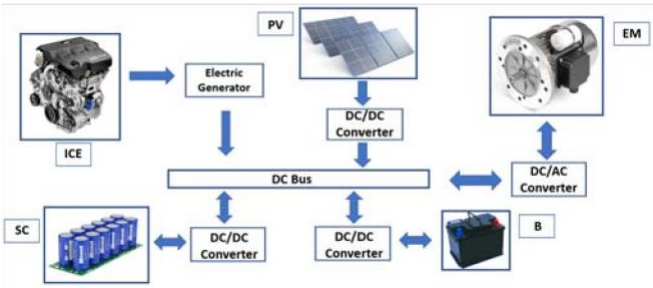

Fig. 4. Example of a series architecture for Hybrid Solar Vehicle.

vehicle, and the tail pipe emissions. Many researches [23][24] have proved that HEVs have the lowest values of "wells-towheels". On the other hand, HEVs may produce a higher wellsto-wheels emissions value in comparison to the combustion ones when they are charged through a power grid with power generators using high polluting fuels. Notwithstanding, considering that the worldwide concerns about environmental safety power production are driving towards an increasingly diffusion of technologies based on renewable energy [2], this will make the power grid become greener and, in turn, the HEVs can be considered as a very low emission transportation means. In this perspective the HSVs have a great potential thanks to the use of renewable energies.

Many countries are supporting programs on electro mobility, *e.g.* the USA government is investing more than $140 billion to have a strong reduction of the pollution emission. It is clear that the pollution, climate change, and HSV are related to each other. More specifically, the solar energy depends on the weather conditions and, in turn, the distribution of the solar irradiation is affected by pollution as described in [25]. In this paper the world surface with an estimation of the difference between the solar map energies for the period 1949–1958 and for the period 1999–2008 are recalled. The comparison highlights that the solar energy at the surface over the United States is decreased.

B. Economic impact

In the design and implementation of a HSV, a crucial point is the evaluation of the economic impact when the PV panels are installed on board of the vehicles. In this regard, Simulation and optimization are useful tools [26][27]. Many researches show that the in some US region the use of a HSV lead to considerable savings on the grid electricity (using on-board PV) around 4.5 to 21.1MWh per EV lifetime as shown in [28] compared to conventional HEV. To evaluate properly the impact of the solar vehicles, it is considered the estimated rate of return on investment for the whole lifetime of the vehicle in case of a plug-in EV, otherwise known as ROI: ( ) *Gain of Investment − Cost of Investment*

$$ROI\,\% = \frac{\text{Gain of Investment} - \text{Cost of Investment}}{\text{Cost of Investment}} \times 100\% \qquad (1)$$

In detail, the Cost of Investment is assumed as the sum of the individual cost of the PV modules, MPPT, the mounting, the installation and maintenance. On the other hand, the Gain on Investment is assumed as the product between the efficiency of the vehicle (Wh/mile), the PV Daily Solar Range, the cost of electricity ($/kWh), the vehicle lifetime in year. The total service of maintenance per year (M/year) [29] cost is assumed to 2% of the total system cost with the assumption of an inflation rate of 3% and an interest rate of 10% [29]. The lifetime of the PV system is considered equal to the life of the vehicle [30]. The total cost of the maintenance is used to calculate the gain of investment, as estimated in [28]. It includes the PV daily solar ranges, the vehicle efficiency referred to the total journey, the vehicle lifetime in year and the cost of electricity. In [28] an interesting analysis referred to the economic impact of the HSV considers the cost of investment into two different scenarios: high price and low-price estimation. The experimental test was performed using different type of cars (5 different models) and the driving tests were performed in the months between December and June, into two different regions in the USA, in details in Boston and Phoenix. For example, considering the on-board PV system of vehicle #5 (*e.g.* Tesla) in Arizona, the Gain on Investment was increased around to 1.0 times the investment cost, while in Boston, having a low solar energy density, a small negative ROI was obtained, around -25%. Many different conditions were evaluated during the six months of tests. The calculations aforementioned are referred to the vehicle lifetime (about 12 years), with the assumption that these PV modules can be reused to other applications after that the vehicle is recycled.

C. Impact on the grid

The advantages of electrical vehicles and HEVs for the smart grid development have been widely investigated, as well as the challenges arising in the management of the electrical network in presence of a large diffusion of dispersed charging points and stations [31], [32]. The HSVs outdo them in supporting the smart grids thanks to their solar panels, indeed they can be considered as mobile distributed generators with a storage system. Similarly, HSVs' impact on the grid is mitigated by means of the on board renewable generation, indeed they can be considered as prosumers. Notwithstanding, there is not any works in literature treating the impact of HSVs on the future electrical networks. Actually, a HEV connected to a charging point, having PV panels, allows estimating the effect of the HSV on the grid. More specifically, the system composed by the HEV and the PV panels of the charging station can be considered as a HSV when the advantages and challenges for the electrical network due to HSVs have to be investigated [33]-[35].

IV. Components of the Hybrid Solar Vehicles

A. Power converters

The key point for the motion in the HSVs is the process of converting the electrical energy into a mechanical energy. Most of the energy from the PV panels is regulated with several DC/DC converters. The DC/DC conversion systems are built with several components. The DC/DC converter (buck, boost, etc.) produces a controlled output dc voltage and it determines the average DC voltage applied to the electrical load. Furthermore, it should be used passive components to realize the filters with high efficiency [36]-[38]. Simulation tools devoted to the conversion engines (PV, power converters, devices, etc.) are useful to estimate the fields of temperature or hot spot in dedicated regions [39]-[44].

In [45] an interleaved boost DC/DC converter is presented aiming to develop a fast dynamic response converter suitable for HSV, more performant than the conventional designs treated in literature. In particular, both a current control loop to minimize the input current ripples and a low output voltage ripples are obtained. A complete modeling and design of each part of an HSV is proposed in [46]. In details the DC/DC converter design provides a dynamic faster than those of both the PV panels and the lithium-ion batteries that are used into the prototype, thus the study of the transient response of the converter can be neglected. Consequently, the model can be simplified and the efficiency of the whole DC/DC converter can be assumed as a function of the input power and irradiance level. All the simulations results are in accordance with the experimental ones, and confirm the validity of the used approach.

A wide variety of converters has been examined in [47], both unidirectional and bidirectional ones. In detail, the DC/DC converter designed for the PV modules is a unidirectional one. it has been chosen a buck-boost converter that it is designed for the MPPT. It converts the fixed DC bus voltage to lower or higher voltages on the PV cell and batteries. The model of the converter has been developed in a Matlab/Simulink environment. Therefore, a control signal with a duty cycle of 0.615 is generated to feed the gate of the IGBT switches. In [48] the DC/DC converters are used as switchingmode regulators to convert the unregulated DC voltage, to a fixed DC output voltage. The used boost converter has an input voltage of 24V and it is boosted up to 48V, and the switching frequency used is 20 kHz. The design of this conversion system require the proper choice of the inductor. In this perspective, some equations are proposed in the paper to set the inductance. It is worth noting that the switching diode used in the converter must withstand a reverse voltage at least equal to the circuit output voltage and must be able to conduct the peak output current: a Schottky barrier diodes is often used. A discontinuous current supplies the output RC load, thus requiring a large capacitance to limit the output voltage ripple.

B. Optimal Energy Management

In literature many works are referred to the study and development of innovative energy management in case of HEVs [49]-[53]. In case of HSVs, different strategies can be used, in fact, the ESS can be recharged during the parking mode: this means that the state of the charge of the battery pack can be maintained constant for the whole day. In [53] a novel rule-based control strategy that aims to optimize the onboard energy management of a HSV is presented. It is implemented using a controller with a processor and userprogrammable FPGA control unit that allow to perform experimental tests on a HSV prototype. The control system is made up by two main loops. The external one estimates the state of charge that should be harvested during the remaining journey, as function of the expected solar contribution. The forecast solar contribution is evaluated using a widely available solar database and by processing past solar energy data measured on the vehicle. Neural Networks predictors, previously stored data and/or GPS derived information are suitable to estimate the average power requested by vehicle for the traction. The internal loop aims to define the optimal power value for the EG and to avoid oscillation. The control strategies have been developed in order to increase the fuel savings in the HSV thanks to an intermittent use of the ICE.

A model predictive control is considered in [54] for a realtime control strategy able at minimizing the fuel consumption in a HSV. An equivalent parameter that takes into account the fuel consumption during the vehicle operation has been defined. The model is used to calculate the amount of useful fuel that must be provided to the battery to reach a properly value of SOC. Different tests have been performed and three parameter configurations were chose, obtaining a fuel consumption minimization for these configurations. The work in [55] deals with a detailed study, control and energy management of a solar vehicle with multisource power supply. The main energy sources are the PV cells and a series of battery storage. Thus, the power sources are connected to the DC bus by two DC/DC converters to establish the power transmission to the traction part. The PV cells are connected to a boost converter that is controlled by an MPPT algorithm. The buck-boost converter connected to the battery array is operated through an energy management strategy able at satisfying the load power requirement. The simulations of the whole system have shown the effectiveness of the control strategy, especially in an urban area.

Reference [56] focus on the development of an innovative strategy to properly manage the storage energy in a HSV. The core of the project is related to the study of the performance achievable by an intermittent use of the ICE that is powered also with an EG. It is worth noting that the authors have studied the series architecture of an HSV. A relevant comparison with the HEV was also carried out. The optimal energy management is obtained thought the application of Genetic Algorithms (GA) [57]. The optimal powering strategy takes into account for fuel mileage and state of charge. It is also taken into account the solar contribution during the parking mode. Otherwise in [58], the work proves the improvements of the performance of a HSV already developed, with an innovative power management system. Solar panels, super capacitors and batteries are combined together. In this application, the super capacitor can provide better dynamic performance to the vehicle. As the main part of the project, to manage the different resources and consequently the power flow of these, a control strategy based on fuzzy control theory was designed. The authors evaluated the efficiency of their proposed power management system, and it has one of the highest efficiency among other works analyzed in literature.

C. Energy Storage System

As well known, a key point in the HSVs is to maximize the range of autonomy during a journey. In [59] the storage system is formed by a combination of supercapacitors and batteries to improve the efficiency of the overall system. The supercapacitor has been proposed as an auxiliary energy storage element. Indeed, the supercapacitors store the energy during fast power charge and it recovers the braking energy to increase the vehicle autonomy. In addition, this work faces the modeling, control and power management of the electric traction system of a HSV using solar energy as primary energy source. The simulation results have highlighted the effectiveness of integrating the supercapacitor in the energy storage system of the vehicle. In detail, it reduces the current peaks in the battery and also the battery life is increased, proving that the electrical stresses on the battery are reduced. An innovative solar power-assisted ESS balancing system in an EVs is proposed in [60]. This work is based on solar power harvesting and storage. During the journey (i.e. during the ESS discharge) the proposed method is designed to charge the ESS with the lowest state of charge during the discharging process through solar irradiance (solar-balancing mode) or using the energy from an auxiliary storage cell (storage-balancing mode). During the parking mode, when the ESS is recharged by plug-in connection the adopted method allows to discharge the cell with the highest state of charge passing its energy in the auxiliary storage cell (charge-balancing mode) instead of waste the energy using a resistor. Simultaneously the PV panel recharges the ESS to maximize the solar energy harvesting. The simulations and experimental results confirm that the system achieves a performance improvement with respect to the conventional balancing systems during both the charge and the discharge of the ESS.

D. PV system

The specific mathematical model representing the generator operation on efficient mode is important when designing the PV generator supplying the EM. Therefore [61] proposes a comparative study and analysis between various models of PV cell installed in an HSV. The model providing the best approximation of the PV cell has been selected thanks to the data sheet parameters. The comparison among the single diode model, two-diode model, and the ideal single diode model has been carried out. The simulation results obtained using each model has been compared to the experimental data of the P-V and I-V curves for various irradiance and temperature levels when using the maximum power point tracking (MPPT) algorithm and the root mean square deviation [62]-[64]. In [65] an artificial neural network(ANN) based MPPT for HSV is proposed. The MPPT is performed into a highly efficient boost converter with IGBTs as power switches [66]. The reference voltage for the MPPT is obtained by an ANN combined with gradient descent momentum algorithm. The tracking algorithm modifies the duty-cycle value of the boost converter in order that the PV-module voltage equals the voltage corresponding to the maximum extracted power. All these assumptions are valid in any irradiance grade, temperature, and load conditions. To increase the speed response, the system is also driven by a digital signal processor. A proportional-integral-derivative (PID) controller that imposes and matches the reference and battery voltage levels is adopted to ensure the system stability. The ANN provides information to the PID controller that sets the boost converter duty-cycle. The energy produced enables the charge of the lithium ion battery on board. The simulation results have shown that the algorithm is highly efficient in comparison with other works in literature.

Reference [67] proposes an innovative MPPT algorithm that it is implemented on a HSV. The algorithm is based on a numerical method with variable step. The performance of this MPPT algorithm, and other ones presented in literature, has been assessed by simulation in Simulink. The results validate the effectiveness of the algorithm and it provides a better way to find the best MPPT, both in terms of speed and accuracy. A MPPT hardware that it is installed on board of a HSV presenting for silicon junction cell applications is designed in [68]. The dimension of the MPPT hardware is 17 mm by 21 mm and it works with an input voltage range from 0.9 V to 1.8 V, having a nominal power rating of 600 mW. It is suited for a power tracking group of four series connected silicon cells. The system presents a power conversion efficiency of 93.9%, which has been evaluated by measuring the power losses through the calorimetric method. A solar cell simulator is also used to evaluate the performance of the MPPT hardware. The experimental results show that the algorithm reaches a high tracking accuracy under both the static and the dynamic irradiance conditions.

In the case of solar electric vehicles moving in a city, it is necessary the adoption of MPPT methods well suited for operating under fast-changing position and partial shading conditions [69].

CONCLUSIONS

This paper reviewed the current status, impacts and opportunities of HSVs deployment, as well as the latest development of the solar technologies. The paper has also summarized the impacts of HSVs roll out on the environment, economy and power grid. The latter one has not provided many results in literature, so it should be studied more in depth in the next future. Moreover, since the current efficiency level of PV panels available in the market is too low, then the advantages deriving from the harvesting solar energy are limited considering the increasing of the payload and costs. Consequently, very high efficiency panels are necessary in view of the limited areas of PV

system than can be installed on the vehicle. Finally, MPPT methods well designed for such mobile application are necessary. In particular, such methods have to be able to cope with PV systems operating under fastchanging positions and partial shading conditions.

REFERENCES

[1] Arsie, I., G. Rizzo, and M. Sorrentino. "A Model for the Optimal Design of a Hybrid Solar Vehicle." Review of Automotive Engineering 29 (2008): 439-447.

[2] Bai, S., D. Liu, and M. Li. "A Review of High Efficiency Photovoltaic Technologies for Vehicle Applications." Renewable Energy 36, no. 11 (2011): 2896-2906.

[3] Bernardes, M., A. Voss, and G. Weinrebe. "Thermal and Technical Analyses of Solar Chimneys." Solar Energy 75 (2003): 511-524.

[4] Boscaino, V., V. Ferraro, R. Miceli, C. Cavallaro, and A. Raciti. "Performance Evaluation of a Multisource Renewable Power Converter Prototype." IEEE PEMC, Antalya, 2014, 1318-1323.

[5] Cabrane, Z., M. Ouassaid, and M. Maaroufi. "Performance Enhancement of Solar Vehicle by Integration of Supercapacitors in the Energy Storage System." IRSEC, Marrakech, 2016, 62-67.

[6] Ceraolo, M., G. Pede, A. Pozio, and A. Vellucci. "A Parallel-Hybrid Drive-Train for Propulsion of a Small Scooter." IEEE Transactions on Power Electronics 21, no. 3 (2006): 768-778.

[7] Chimento, F., A. Raciti, and S. Musumeci. "A Control Algorithm for Power Converters in the Field of Photovoltaic Applications." European Conference on Power Electronics and Applications, Aalborg, 2007, 1-9.

[8] Clemens, L., Mart√≠nez-Botas, N., Howey, Lytton, and Cussons. "Comparative Analysis of the Energy Consumption and CO2 Emissions of 40 Electrics, Plug-in Hybrid Electric." Hybrid Electric and Internal Combustion Engine Vehicles 23 (2013): 12-19.

[9] Dayem, A. M. A. "Set-up and Performance Investigation of an Innovative Solar Vehicle." 1st International Workshop on Diffusion of New Energy Technologies in China, Shanghai, 2011.

[10] Duan, C., H. Bai, W. Guo and Z. Nie. "Design of a 2.5-kW 400/12-V High-Efficiency DC/DC Converter Using a Novel Synchronous Rectification Control for Electric Vehicles." IEEE Transactions on Transportation Electrification 1 (2015): 106-114.

[11] Egiziano, L., A. Giustiniani, and G. Lisi. "Experimental Characterization of the Photovoltaic Generator for a Hybrid Solar Vehicle." IEEE Int. Symposium on Industrial Electronics, Vigo, 2007.

[12] ElMenshawy, M., M. ElMenshawy, A. Massoud, and A. Gastli. "Solar Car Efficient Power Converters' Design." IEEE Symposium on Computer Applications & Industrial Electronics, Penang, 2016, 177-182.

[13] Garrison, M. "The 2007 Solar D House." 5th International Conference on Urban Regeneration and Sustainability, Skiathos, 2008.

[14] Gibson, T. L. and N. A. Kelly. "Solar Photovoltaic Charging of Lithium-Ion Batteries." Journal of Power Sources 195 (2010): 3928-3932.

[15] Gope, R., A. Mitra, and K. Goswami. "Congestion Mitigation Considering Solar Electric Vehicle: A Possible Solution for Today's Electricity Market." Int. Conf. on Tech. Advancements in Power and Energy, Kollam, 2014, 1-6.

[16] Hagman, J., S. Ritz√©n, J. J. Stier, and Y. Susilo. "Total Cost of Ownership and its Potential Implications for Battery Electric Vehicle Diffusion." Research in Transportation Business & Management 18 (2016): 11-17.

[17] Holman, J.P. Experimental Methods for Engineers. 6th ed. New York: McGraw-Hill, 1994.

[18] Incropera, F. P., and D. P. DeWitt. Fundamentals of Heat and Mass Transfer. 5th ed. New York: John Wiley & Sons, 2002.

[19] Khoucha, F., A. Benrabah, O. Herizi, A. Kheloui, and M. E. H. Benbouzid. "An Improved MPPT Interleaved Boost Converter for Solar Electric Vehicle Application." Int. Conf. on Power Engineering, Energy and Electrical Drives, Istanbul, 2013, 1076-1081.

[20] Klein, S., and G. Nellis. Thermodynamics. Cambridge: Cambridge University Press, 2012.

[21] Koyuncu, T., K.E. Engin, and F. Lule. "Design, Manufacture and Test of a Solar Powered Minibus (Commagene) for Transportation of People in Adiyaman City, Turkey." 29th European Photovoltaic Solar Energy Conference, Amsterdam, 2014.

[22] Letendre, S., R. Perez, and Christy Herig. "Vehicle Integrated PV: a Clean and Secure Fuel for Hybrid Electric Vehicles." Proc. Ann. Meeting of American Solar Energy Soc., Austin, TX, 2003.

[23] Liu, Q. "Electric Car with Solar and Wind Energy May Change the Environment and Economy: A Tool for Utilizing the Renewable Energy Resource." Earth's Future, 2013.

[24] Messenger, R., and J. Ventre. Photovoltaic Systems Engineering. Boca Raton: CRC Press, 2010.

[25] Moran, M. J., and H. N. Shapiro. Fundamentals of Engineering Thermodynamics. 6th ed. Hoboken: Wiley, 2010.

[26] Nanaki, E. A., and C. J. Koroneos. "Comparative Economic and Environmental Analysis of Conventional, Hybrid, and Electric Vehicles: The Case Study of Greece." Journal of Cleaner Production 53 (2013): 261-266.

[27] Poullikkas, A. "Sustainable Options for Electric Vehicle Technologies." Renewable and Sustainable Energy Reviews 41 (2015): 1277-1287.

[28] Rizzo, G., M. Sorrentino, C. Speltino, I. Arsie, G. Fiengo, and F. Vasca. "Converting Conventional Cars in Mild Hybrid Solar Vehicles." IFAC Proceedings 44, no. 1 (2011): 9715-9720.

[29] Sorrentino, M., G. Rizzo, and I. Arsie. "Analysis of a Rule-Based Control Strategy for Onboard Energy Management of Series Hybrid Vehicles." Control Engineering Practice 19 (2011): 1433-1441.

[30] R. Messenger, J. Ventre, "Photovoltaic Systems Engineering", CRC Press, Taylor & Francis, Group: Boca Raton, FL, USA, 2010.

[31] National Household Travel Survey, U.S. Department of Transportation Federal Highway Administration, [Online]. Available: http://nhts.ornl.gov/2009/pub/stt.pdf. Accessed on 27 August 2018.

[32] M. Yilmaz and P. T. Krein, "Review of benefits and challenges of vehicle-to-grid technology," ECCE, Raleigh, NC, 2012, pp. 3082-3089.

[33] G. A. Putrus, et al., "Impact of electric vehicles on power distribution networks," 2009 IEEE Vehicle Power and Propulsion Conference, Dearborn, MI, 2009, pp. 827-831.

[34] J. Robinson, G. Brase, W. Griswold, C. Jackson, L. Erickson "Business Models for Solar Powered Charging Stations to Develop Infrastructure for Electric Vehicles", Sustainability 2014, vol. 6, pp. 7358-7387.

[35] S. Lee, S. Iyengar, D. Irwin and P. Shenoy, "Shared solar-powered EV charging stations: Feasibility and benefits," Seventh International Green and Sustainable Computing Conference, Hangzhou, 2016, pp. 1-8.

[36] J. Tulpule, V. Marano, S. Yurkovich, G. Rizzoni, "Economic and environmental impacts of a PV powered workplace parking garage charging station", Applied Energy, vol. 108, 2013, pp. 323-332.

[37] S. Dadjo Tavakoli et al, "Non-linear time-delay controller for DC/DC power converters in application of electric vehicles",2017, IET Electrical Systems in Transportation, vol. 7, pp. 303-309.

[38] C. Duan, H. Bai, W. Guo and Z. Nie, "Design of a 2.5-kW 400/12-V High-Efficiency DC/DC Converter Using a Novel Synchronous Rectification Control for Electric Vehicles", 2015, IEEE Transactions on Transportation Electrification, vol. 1, pp. 106-114.

[39] A. Raciti, et al., "Electrothermal PSpice Modeling and Simulation of Power Modules," in IEEE Transactions on Industrial Electronics, vol. 62, no. 10, pp. 6260-6271, Oct. 2015.

[40] G. Greco, G. Vinci, A. Raciti and D. Cristaldi, "Generation of electrothermal models of integrated power electronics modules using a novel synthesis technique," THERMINIC, Berlin, 2013, pp. 216-221.

[41] G. Greco, G. Vinci, G. Bazzano, A. Raciti and D. Cristaldi, "Electrothermal model of Integrated Power Electronics Modules based on an innovative layered approach," IEEE IECON, Vienna, 2013, pp. 712-717.

[42] G. Greco, G. Vinci, G. Bazzano, A. Raciti and D. Cristaldi, "Layered electro-thermal model of high-end integrated power electronics modules with IGBTs," IEEE IECON, Dallas, TX, 2014, pp. 1575-1580.

[43] A. Raciti and D. Cristaldi, "Thermal modeling of integrated power electronic modules by a lumped-parameter circuit approach," AEIT Annual Conference, Mondello, 2013, pp. 1-6.

[44] A. Raciti, D. Cristaldi, G. Greco, G. Vinci and G. Bazzano, "Integrated power electronics modules: Electro-thermal modeling flow and stress conditions overview," AEIT Annual Conference, Trieste, 2014, pp. 1-6.

[45] F. Khoucha, A. Benrabah, O. Herizi, A. Kheloui and M. E. H. Benbouzid, "An improved MPPT interleaved boost converter for solar electric vehicle application" 2013, Int. Conf. on Power Engineering, Energy and Electrical Drives, Istanbul, pp. 1076-1081.

[46] S. De Pinto et al., "Electric Vehicle Driving Range Extension Using Photovoltaic Panels", 2016, IEEE Vehicle Power and Propulsion Conference (VPPC), Hangzhou, pp. 1-6.